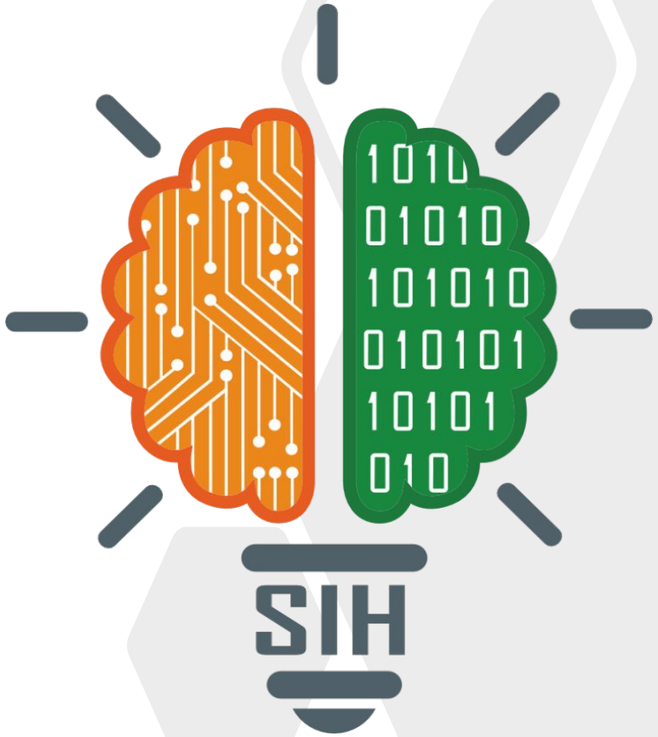
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- **Problem Statement ID –** 25195
- **Problem Statement Title-** Smart Helpdesk

  Ticketing Solution for IT Services
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 100378
- **Team Name (Registered on portal)-** StarWeb



The Challenge: Existing Systems

POWERGRID's existing ticketing infrastructure operates in silos, creating inefficiencies and blind spots.

Multiple Platforms:

Employees use distinct portals (GLPI, Solman, Email) for different issues.

Fragmented Experience:

No single view; employees must navigate complex silos.

Manual Bottlenecks: High reliance on manual sorting and routing of tickets.

Low Satisfaction:

Delays in resolution due to improper routing and lack of self-service.

Key Problem: No common ticket ID, no shared knowledge base, and zero automated analysis across platforms

Proposed Solution: Smart & Centralised

Unified Data Ingestion

Data from various sources flows into a central layer.

Self Resolution of FAQ and Knowledge Base

The system tries to resolve the ticket using knowledge base

NLP Classification

NLP analyzes the ticket to assign to the concerned team



Temporary Ticket Creation

A temporary ticket is generated to capture the request.

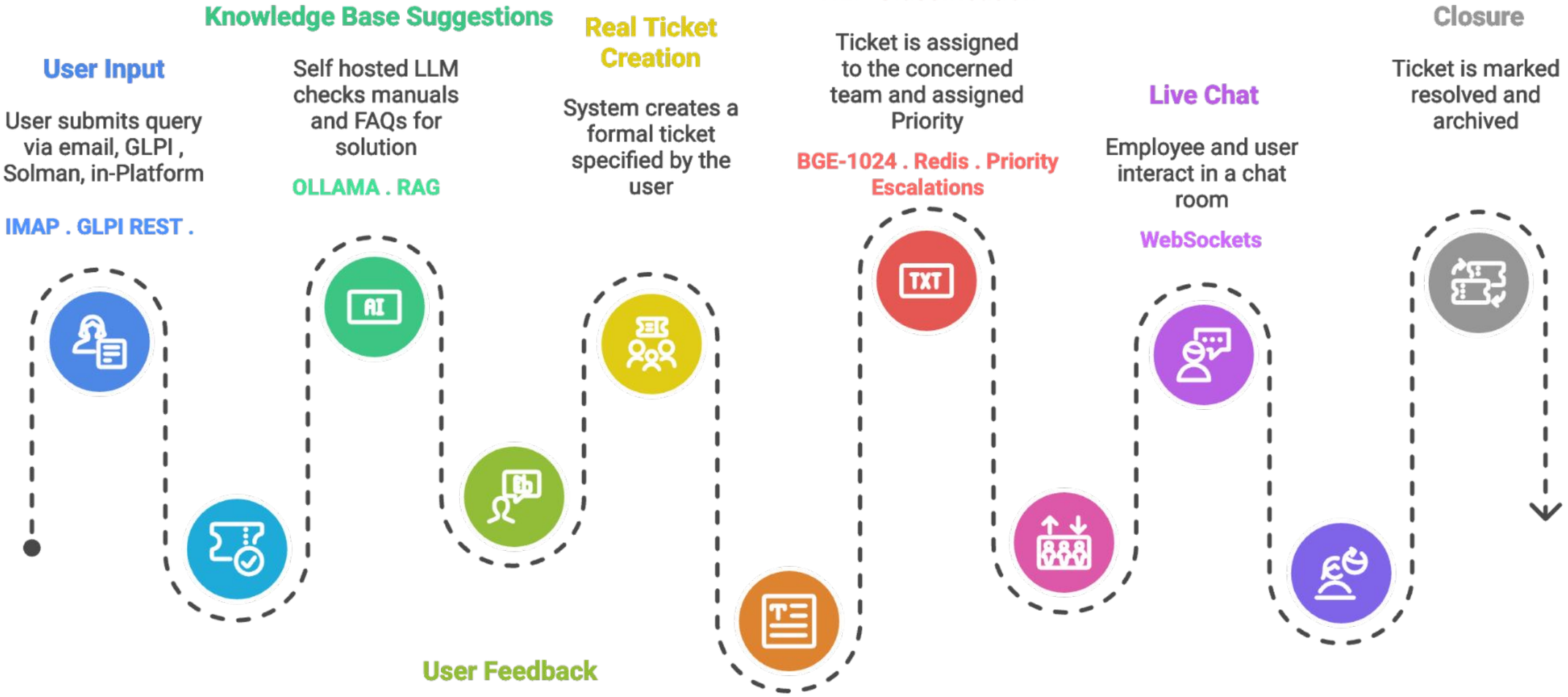
Formal Ticket Creation

If unresolved, a formal ticket is created.

Intelligent Routing

The ticket is assigned to an employee based on workload.

Workflow


User Input
User submits query via email, GLPI , Solman, in-Platform
IMAP . GLPI REST .
Temp Ticket Creation
System generates a temporary ticket
Knowledge Base Suggestions
Self hosted LLM checks manuals and FAQs for solution
OLLAMA . RAG
User Feedback
User provides feedback on AI solution
Real Ticket Creation
System creates a formal ticket specified by the user
StopWords . Wordnet . Semantics
Vector Embedding
Ticket Text is Vectorised
NLP Classification
Ticket is assigned to the concerned team and assigned Priority
BGE-1024 . Redis . Priority Escalations
Intelligent Routing
Ticket is assigned automatically to an Agent
Live Chat
Employee and user interact in a chat room
WebSockets
Resolution
Issue is fixed through interaction
Closure
Ticket is marked resolved and archived

Innovation That Sets Us Apart

Self-Hosted Intelligence

Ollama model fine-tuned specifically on POWERGRID's technical documentation, ensuring data privacy and domain accuracy

Domain-Specific Responses

AI trained on internal knowledge base delivers contextually relevant solutions

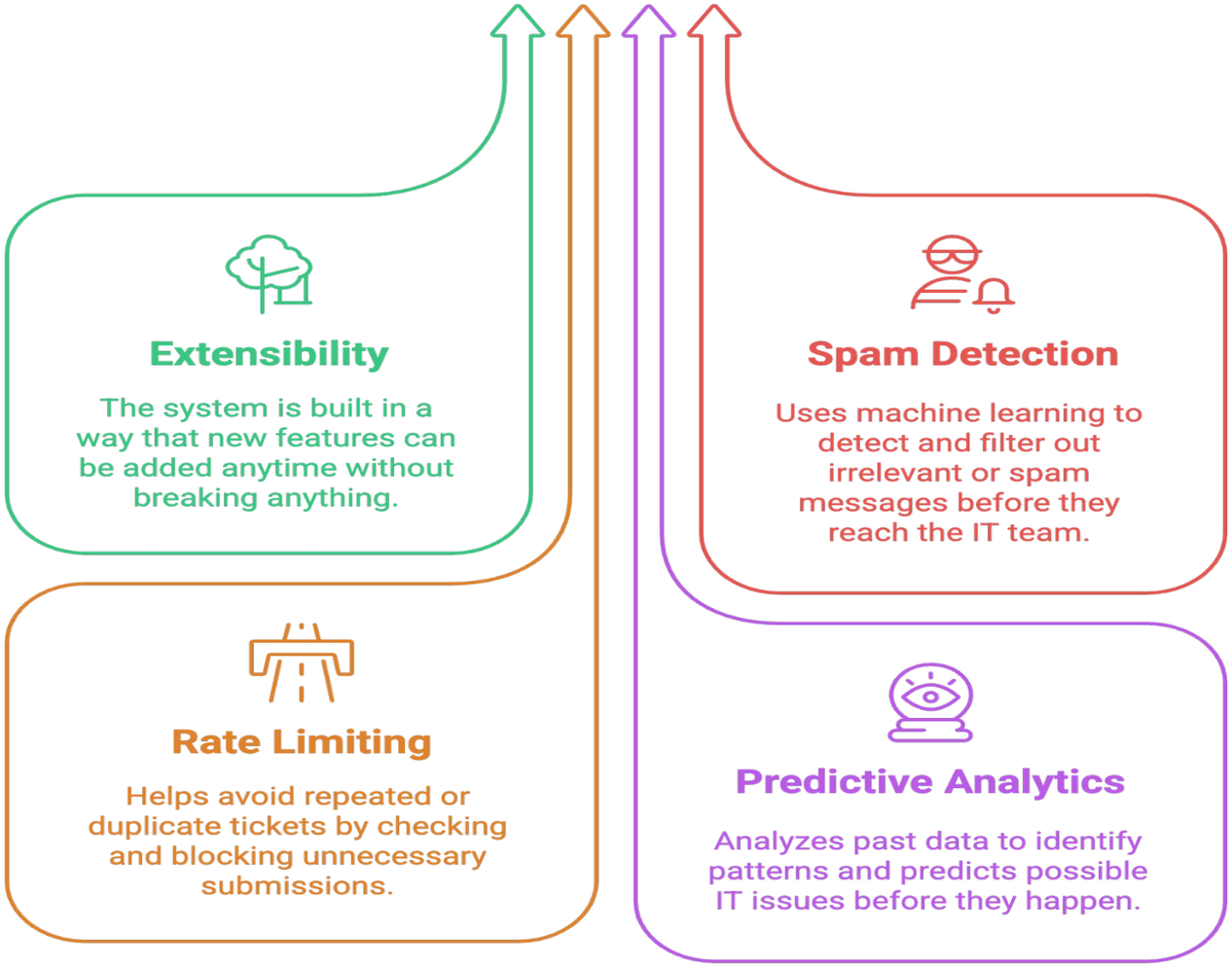
Priority Escalation Engine

Automatically escalates tickets approaching SLA breach, with team leader notifications via email and SMS

Workload Automation

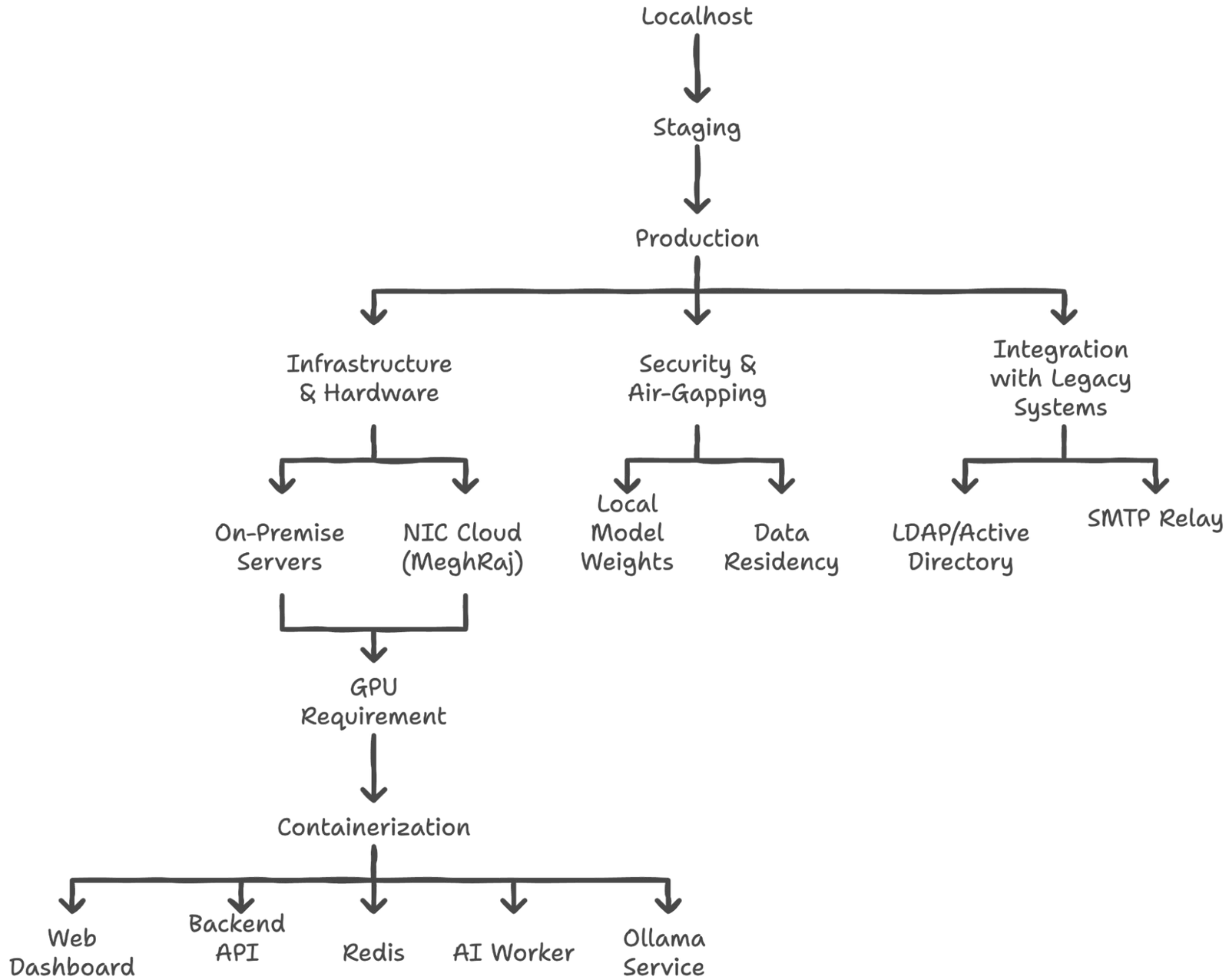
Auto-classification eliminates 70% of manual triage effort, freeing teams for complex problem-solving

Future Roadmap: Continuous Evolution



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Implementation Plan and Strategy



1. **Decoupled Microservice:** AI core runs as an isolated Python container, integrating with legacy Java/Spring ERPs via standard REST APIs.
2. **On-Premise & Secure:** Local LLM inference (Ollama) ensures zero data leakage; capable of running in an air-gapped environment.
3. **Resilient Architecture:** Asynchronous Redis queuing handles massive traffic spikes (e.g., during grid outages) without system failure.
4. **Gov-Cloud Ready:** Dockerized stack compatible with NIC Cloud (MeghRaj) and supports existing LDAP/Active Directory authentication.

Impact

1. Operational Efficiency

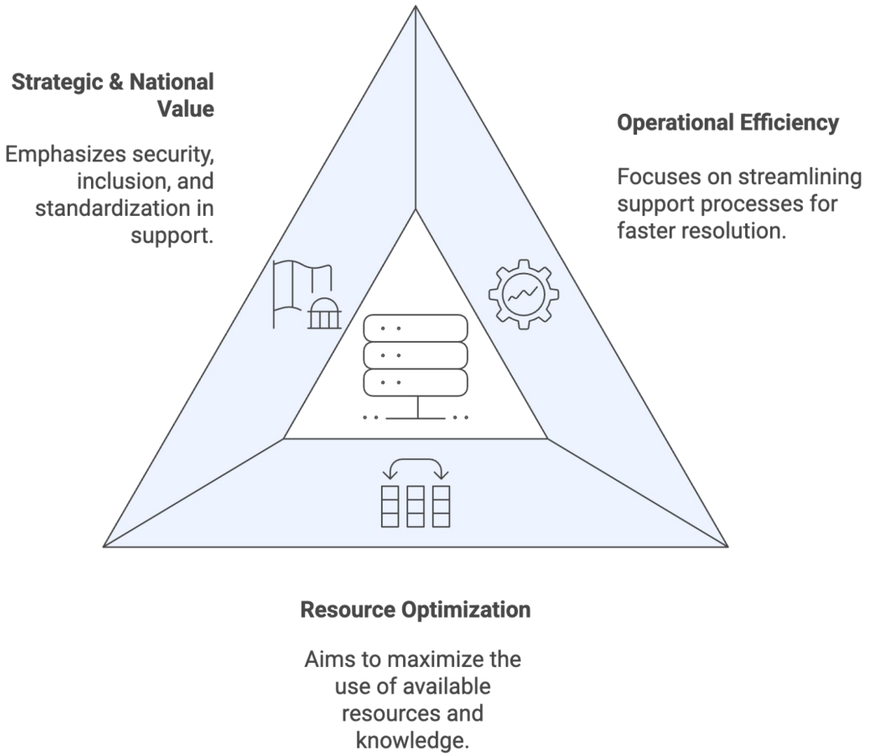
- **Instant Routing:** Eliminates manual triage delays by instantly routing tickets to the correct technical team.
- **Proactive Deflection:** Automates the resolution of routine queries (FAQs, password resets) before they reach human agents.
- **Crisis Resilience:** Prevents helpdesk overload during critical outages by intelligently clustering related incidents.

2. Resource Optimization

- **Workforce Productivity:** Frees up senior engineers to focus on critical grid infrastructure rather than administrative IT support.
- **Automated Knowledge Growth:** The system self-learns from resolved tickets, continuously updating the internal knowledge base without manual effort.
- **Redundancy Elimination:** Automatically detects and merges duplicate tickets, streamlining the support queue.

3. Strategic & National Value

- **Data Sovereignty:** Fully on-premise, air-gapped architecture ensures critical Ministry data remains secure and compliant.
- **Digital Inclusion:** Multilingual support empowers field staff across diverse regions to access IT support in their native language.
- **Standardized Support:** delivery of consistent, high-quality support responses across all departments and regions.



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THANK YOU



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